

Road Accident Hotspot Detection & Risk Prediction

Transitioning urban safety from reactive manual analysis to proactive AI foresight.

Back:

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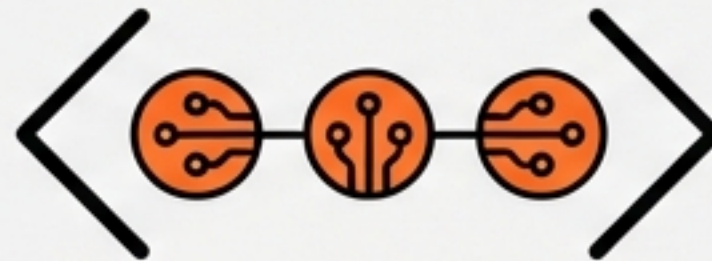
Executive Summary



The Challenge: High accident rates in urban areas coupled with a lack of predictive systems.



The Solution: An AI-ecosystem that detects spatial hotspots and predicts risk scores using historical data.

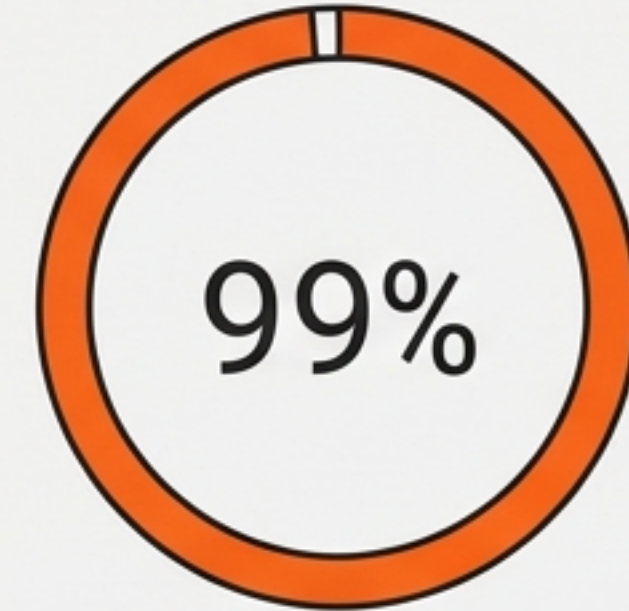
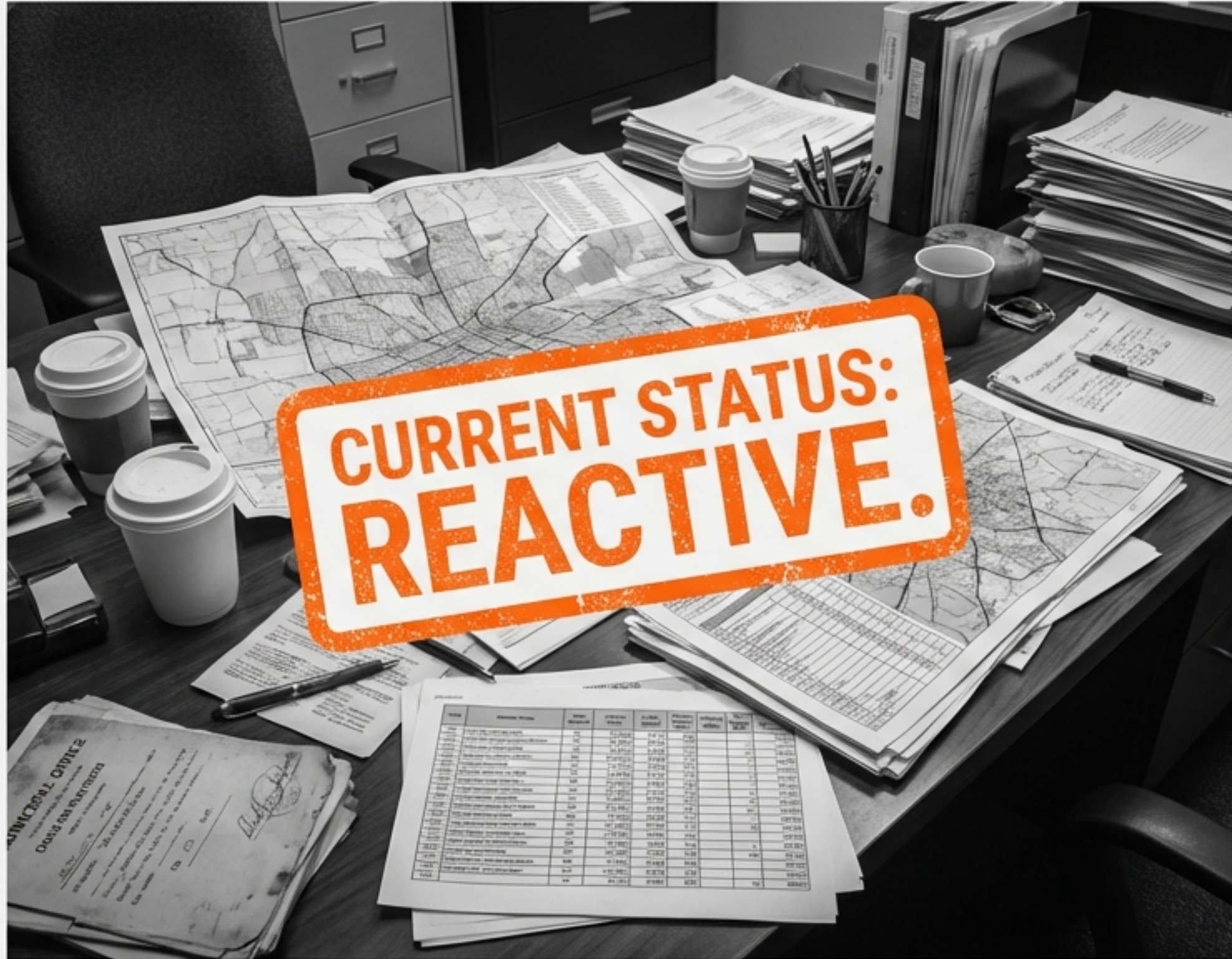


The Core Tech: Unsupervised Clustering (Hotspots) + Supervised Regression (Risk) + Interactive Dashboard.



The Outcome: Data-driven preventive planning for traffic authorities and improved public safety.

The Reactive Trap of Manual Analysis



❗ Connection Timed Out

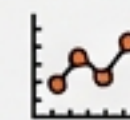
Authorities currently rely on historical and manual analysis. This approach is retrospective—investigating accidents only after they have occurred.



Focus limited to post-event reporting.



Inability to process complex variables for prediction.



Crucial time lag between accident trends and safety interventions.

A Proactive Approach to Risk

We propose a shift from “**What happened?**” to “**What might happen?**”. Our system leverages AI to identify accident-prone zones before they become statistics.



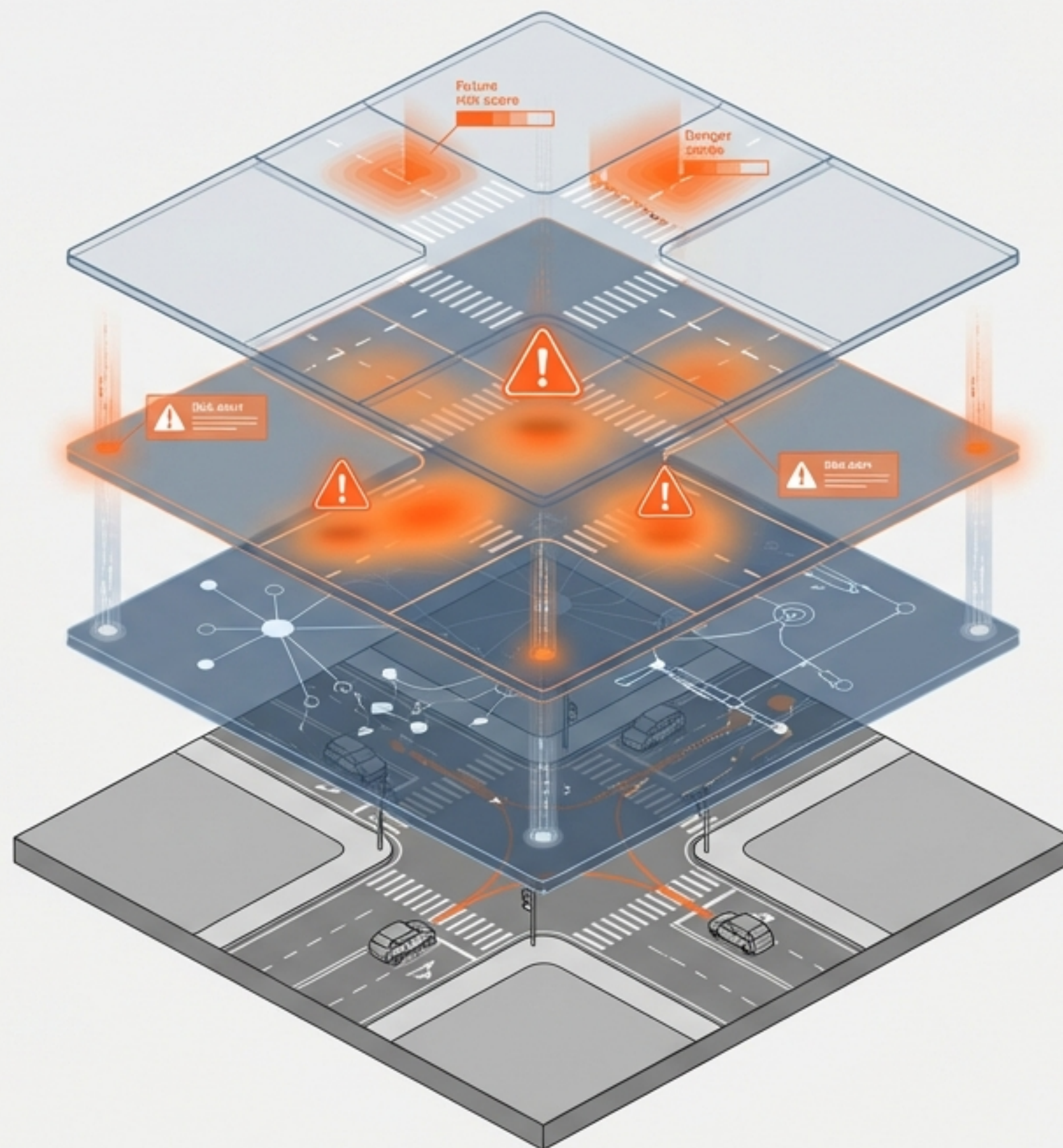
Detect: Pinpoint high-density accident zones.



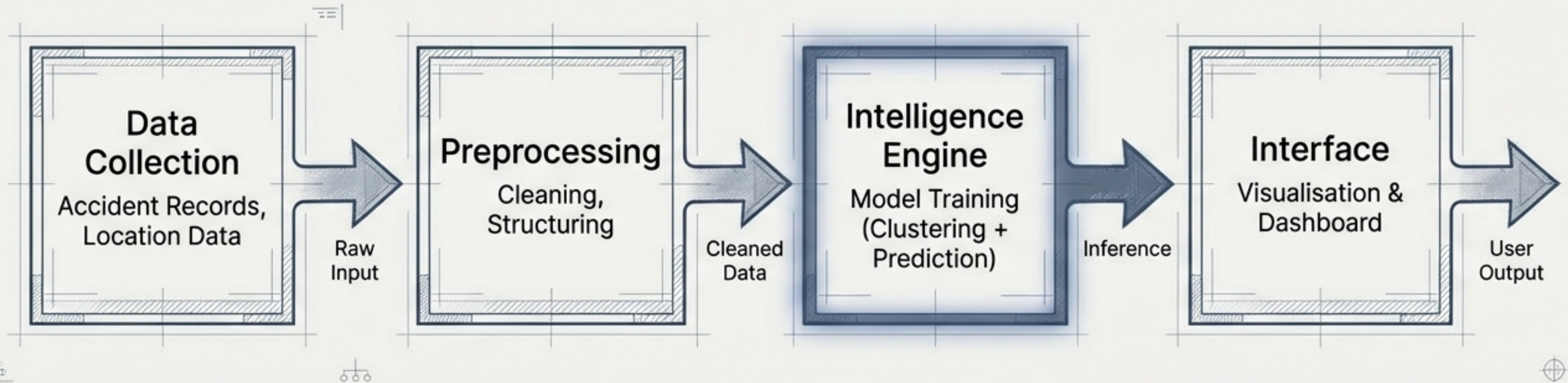
Predict: Calculate risk probability based on environmental variables.



Visualise: Deliver actionable insights via heatmaps.



System Architecture Overview



The Data Foundation

Input Sources



Accident Records: Historical logs of past incidents.



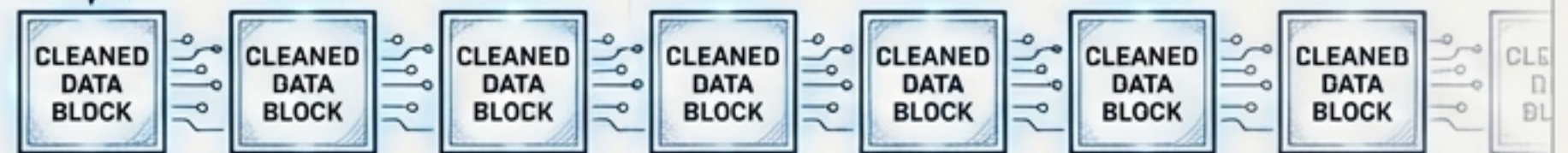
Location Data: Geospatial coordinates (latitude/longitude) of urban and semi-urban areas.



Process



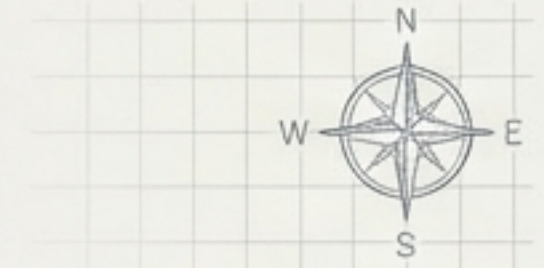
Rigorous Data Preprocessing and Cleaning to remove noise and ensure model readiness.



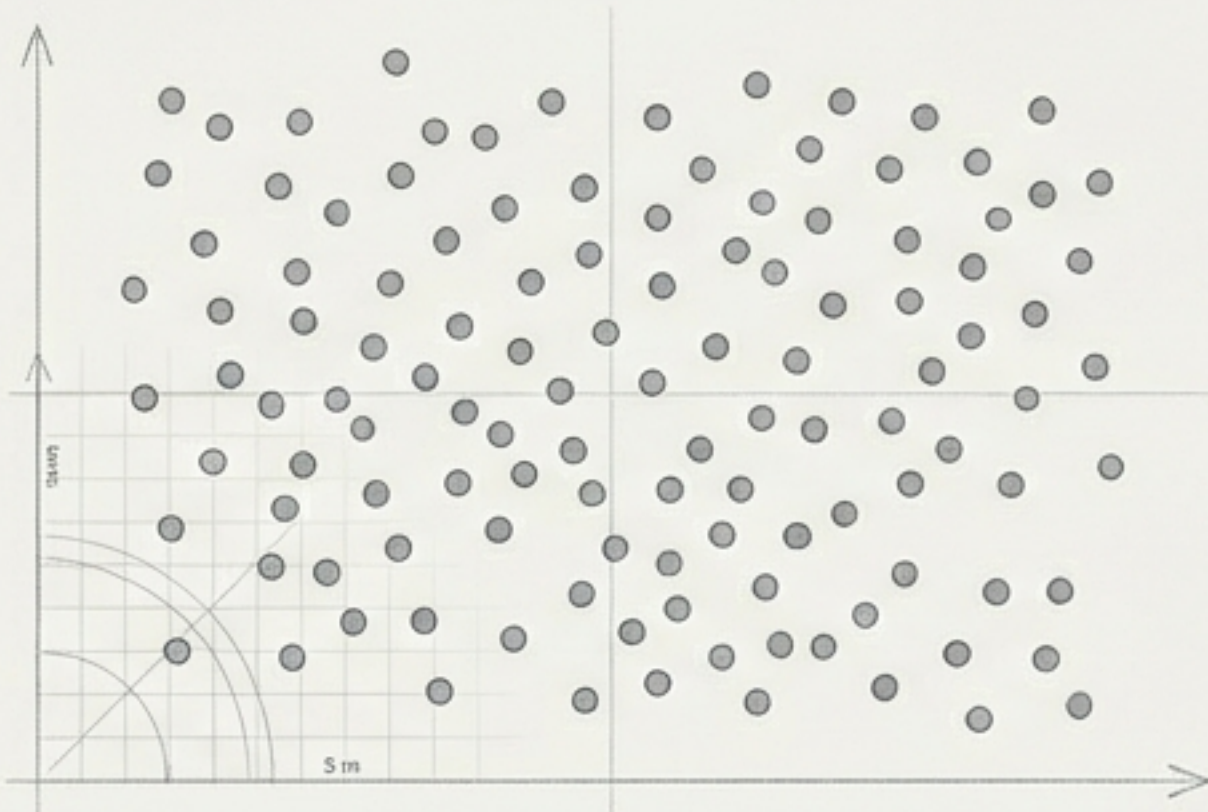
Structured Data for Model Readiness

Mechanism I: Hotspot Detection

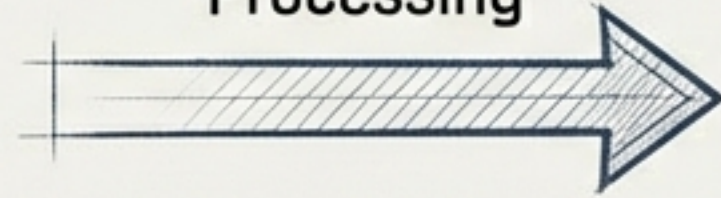
Tech Highlight: Unsupervised Learning



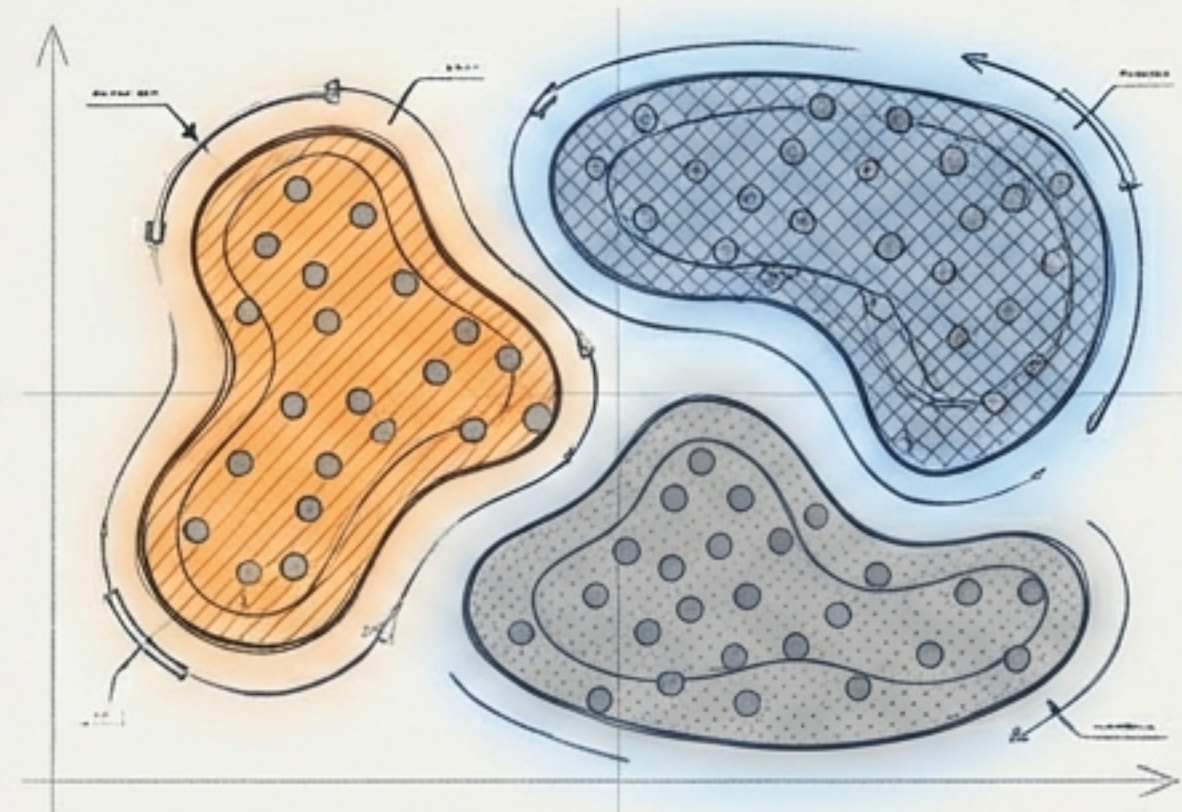
Raw Spatial Data



Clustering Algorithm
Processing



Identified Hotspots & Patterns



Algorithm: Utilising Clustering algorithms such as K-Means or DBSCAN.

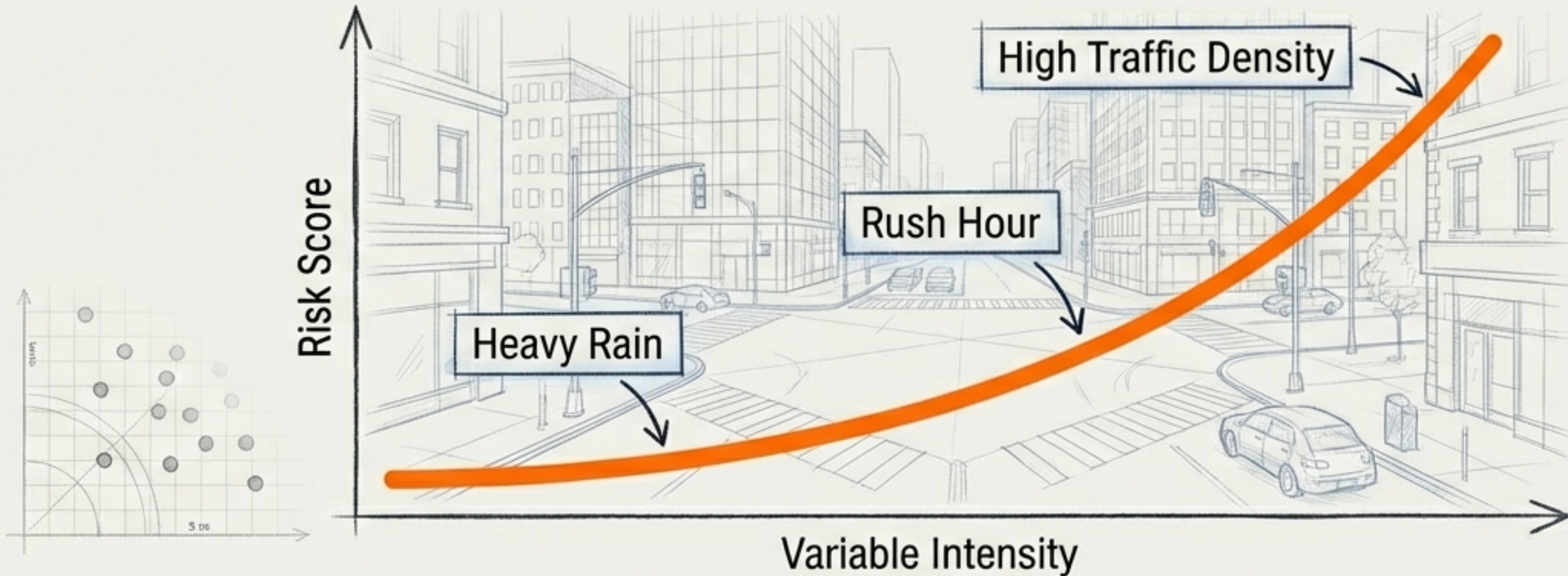


Function: The system aggregates spatial data points to mathematically define 'accident-prone zones,' removing human bias from the identification process.



Mechanism II: Risk Prediction

Tech Highlight: Supervised Learning



Algorithm: Deployment of Regression and Classification models.

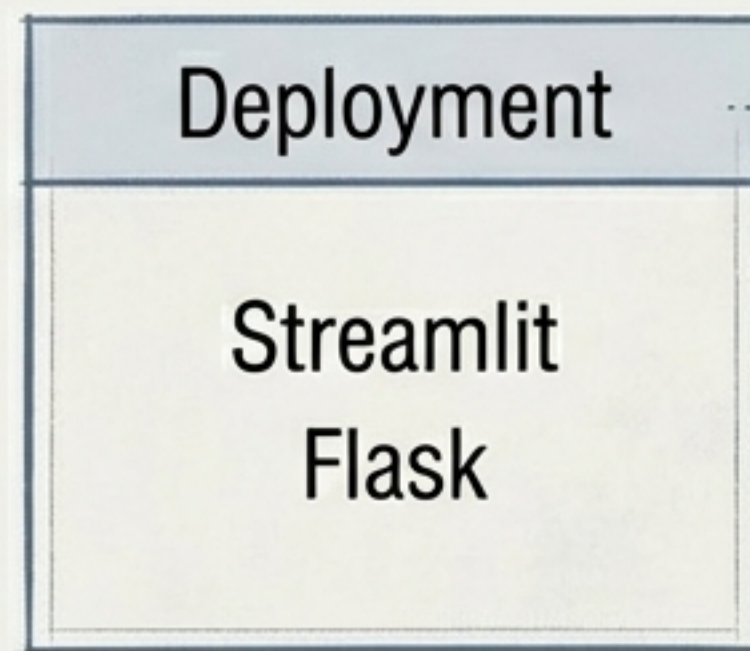
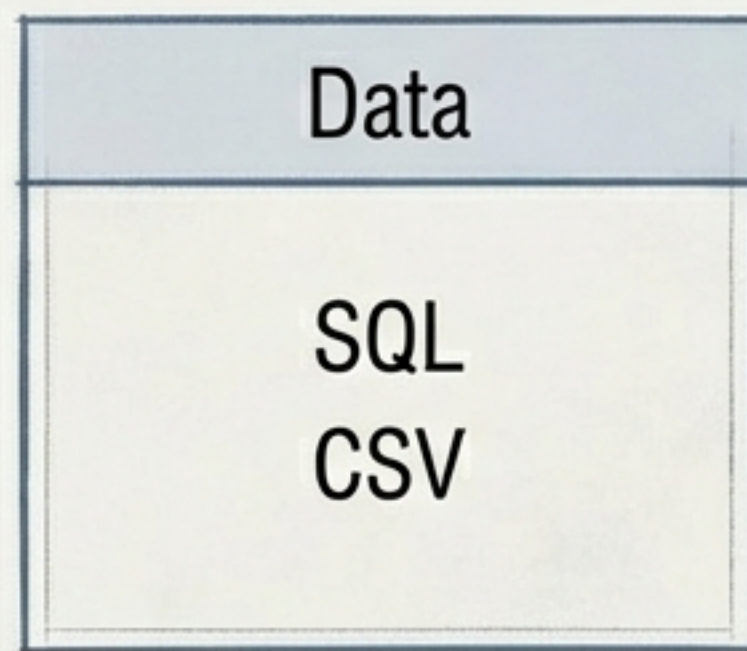
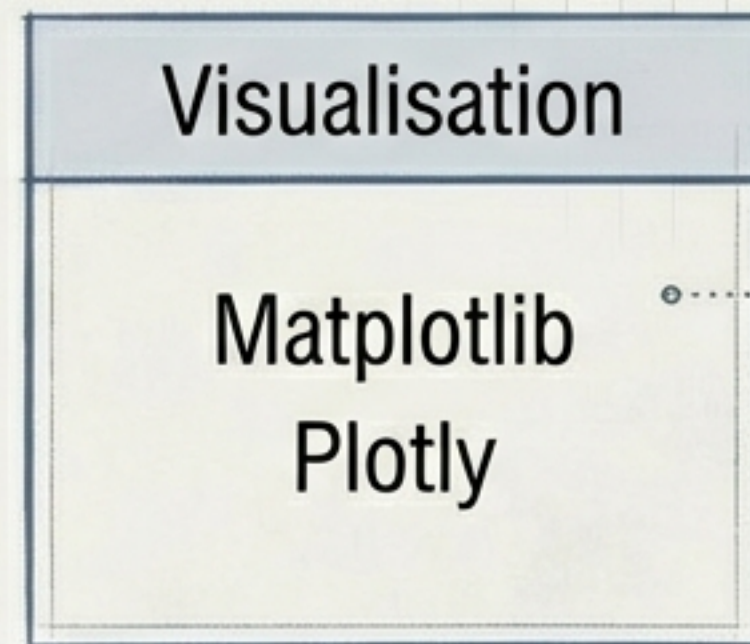
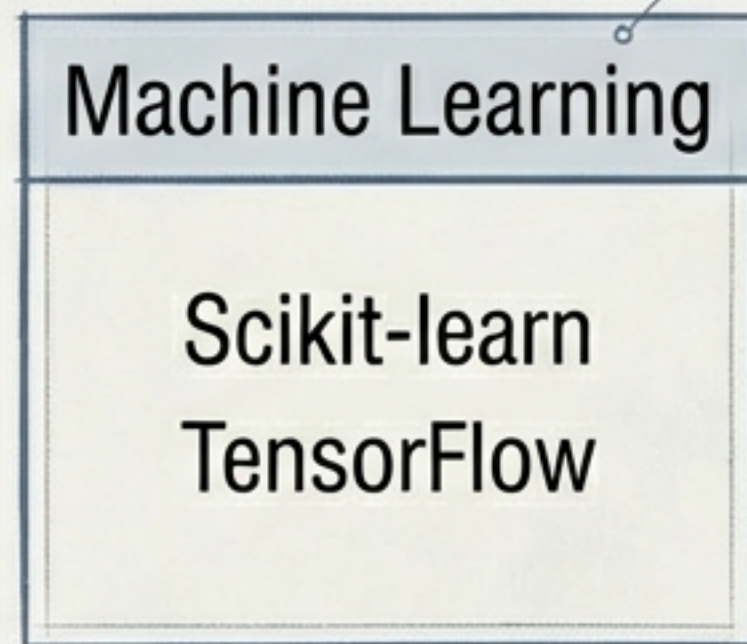
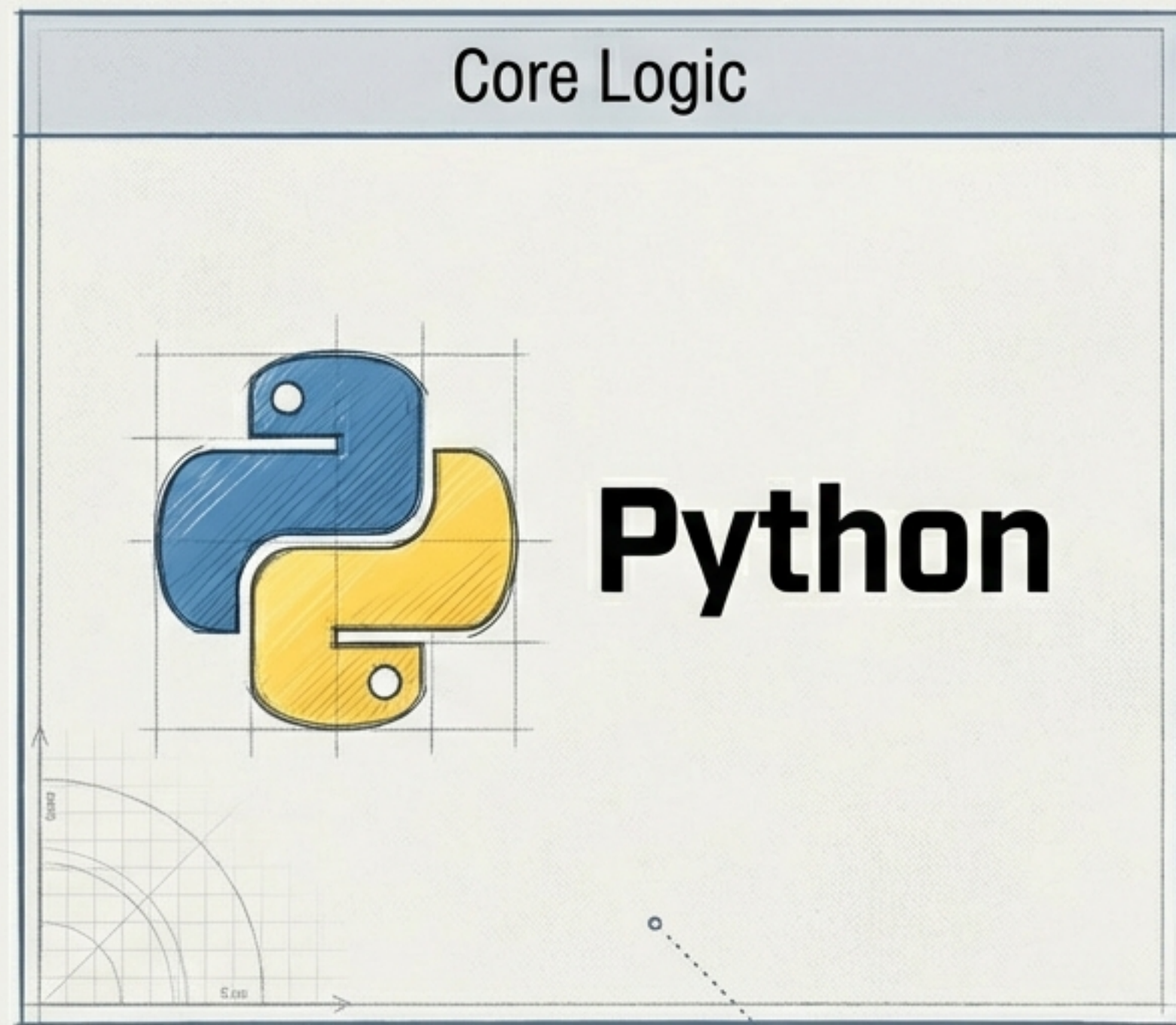


Feature Engineering: The model analyses specific variables to calculate risk scores:

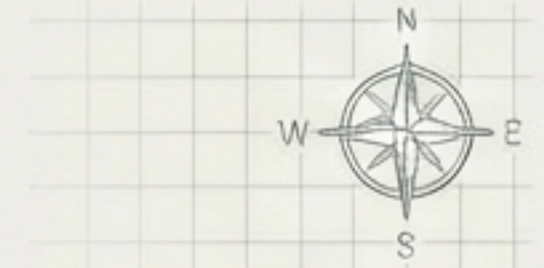
- Weather conditions
- Time of day
- Traffic density



Technology Stack



The User Experience

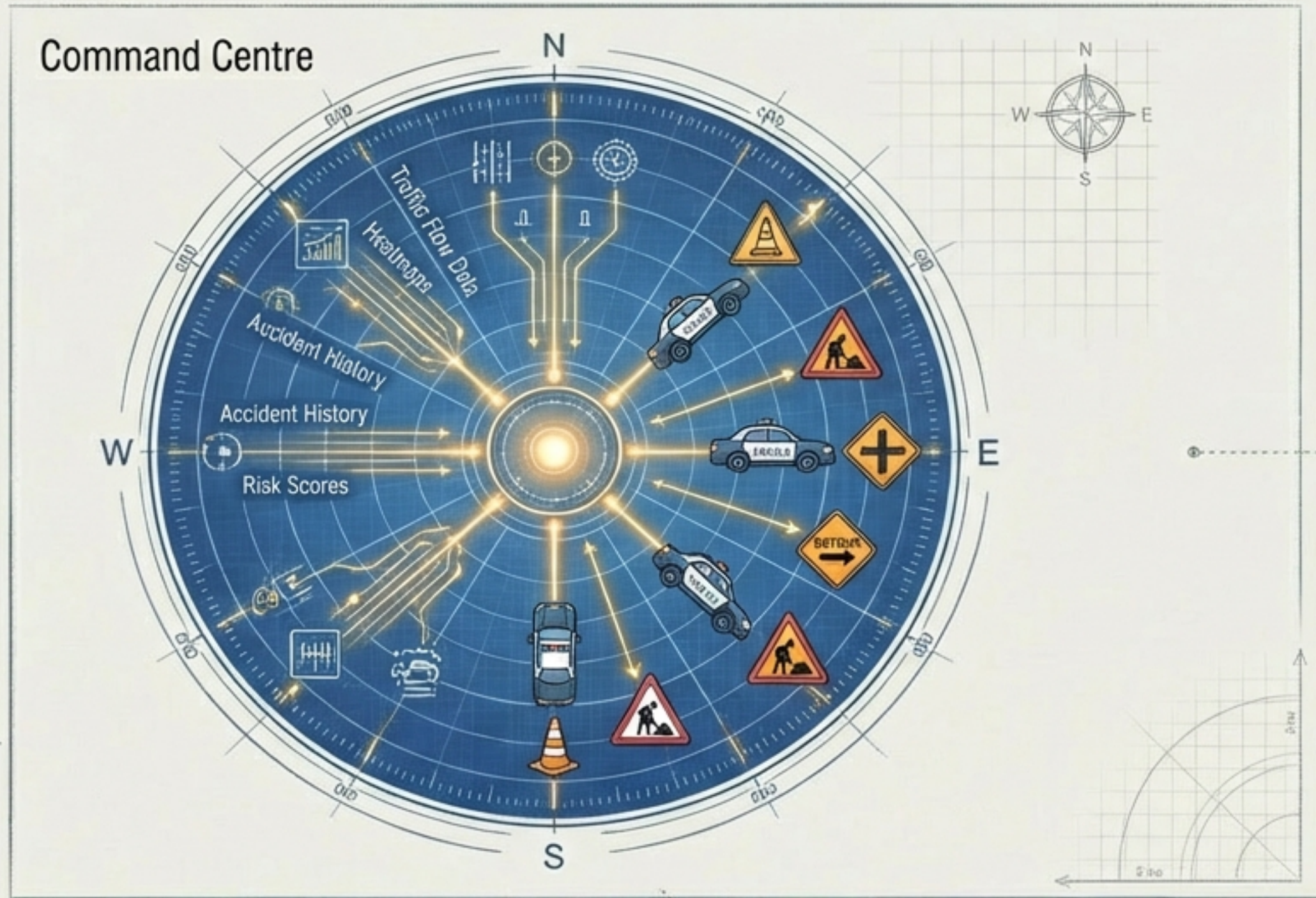


- **Feature: Interactive Map Visualisation**

Authorities access a Web Dashboard (built on Streamlit/Flask) that renders complex data as intuitive visual heatmaps and risk scores, **allowing for immediate situational awareness.**



Empowering Traffic Authorities



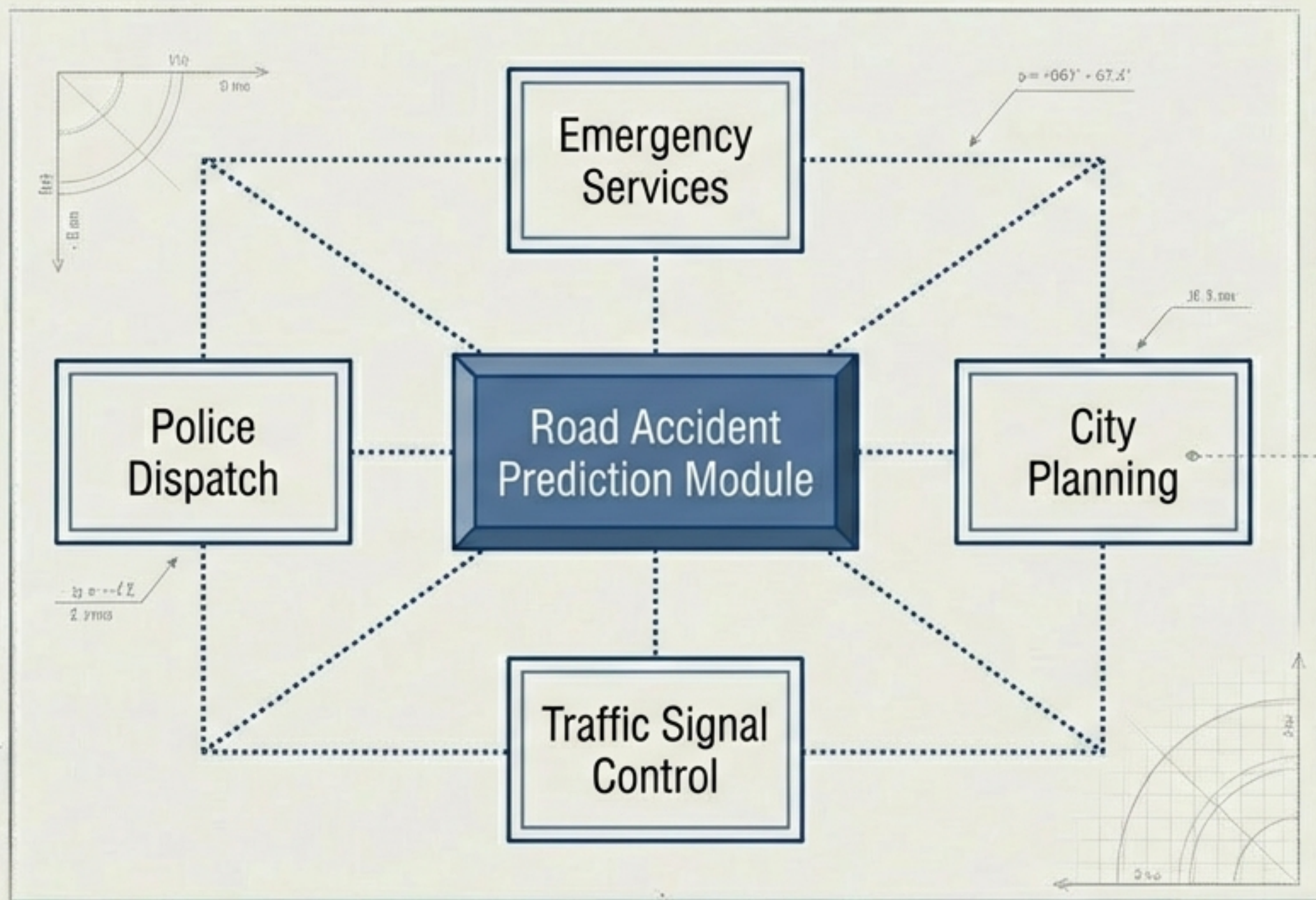
- Supports **data-backed decision-making** rather than intuition.
- Enables **preventive planning**—allocating resources (police, signage, road repairs) to **high-risk zones** before accidents spike.

Societal Impact



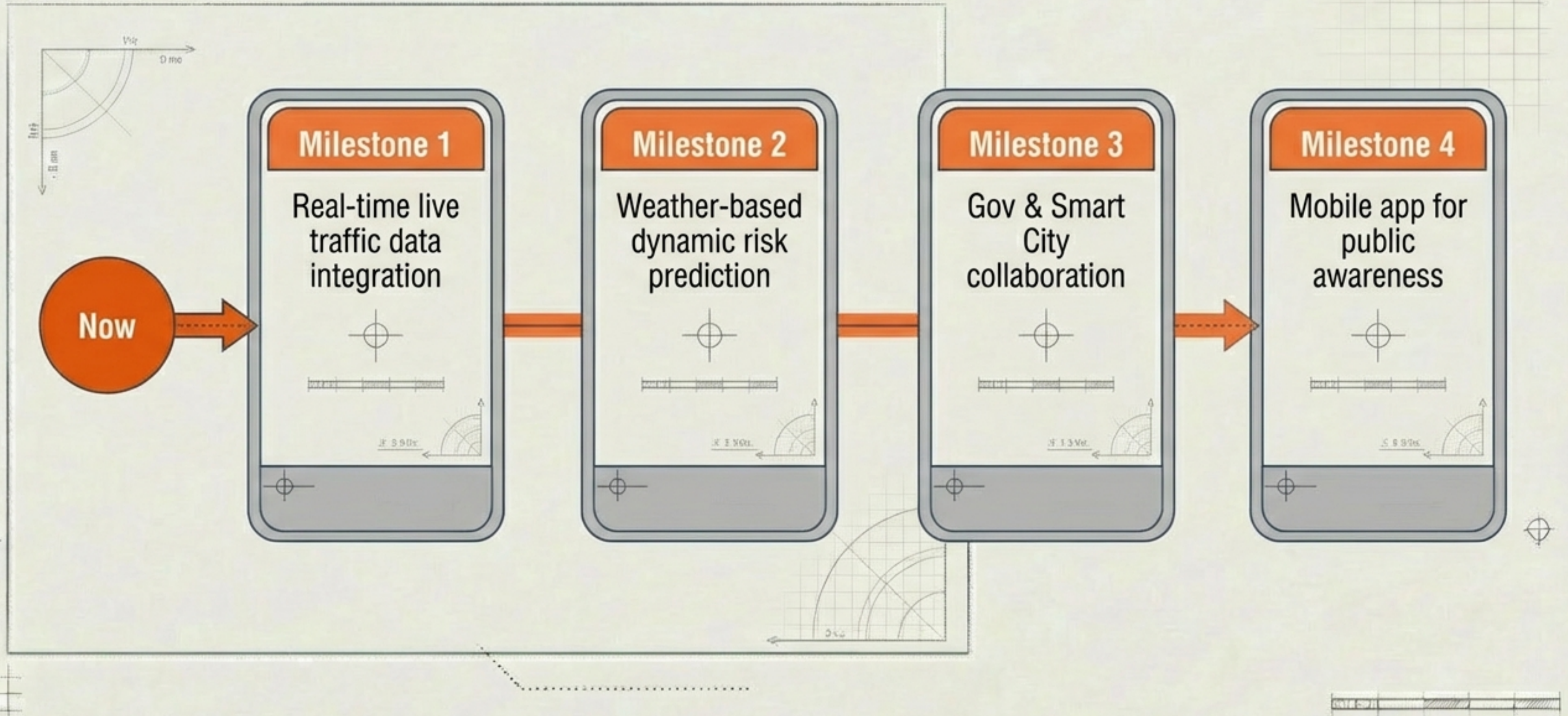
- **Core Goal:** Reduces accident rates through timely intervention.
- **Outcome:** Directly improves public safety in urban environments, creating a safer ecosystem for drivers and pedestrians alike.





- **Integration:** The architecture is designed to be scalable for smart city integration.
- **Flexibility:** The system is not a standalone silo but a module that can feed into broader municipal management platforms.

Future Scope & Roadmap





FROM HINDSIGHT TO FORESIGHT.

Improving urban safety through
intelligent prediction.

Created by Team BACK